AgriSoko Pocket

Group G06 — Technical Report

Mobile application with an AI-powered text-matching service

Group members

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Prepared for academic submission. Confirm all project details and complete the outstanding evidence before submitting.

1. Project Overview

AgriSoko Pocket G06 is a group project designed to help users search team-member information through a mobile interface backed by a text-matching service.

2. Objectives

• Provide a mobile search interface.
• Compare user queries with approved member information.
• Return a match when the similarity is sufficient.
• Avoid treating similarity as proof that a result is correct.
• Validate backend behaviour using automated tests.

3. System Architecture

The project uses a React Native application built with Expo and a Python FastAPI backend. The mobile application sends search requests to the backend. The backend provides GET /health, GET /model-info, and POST /search.

4. AI Matching Approach

The matching service uses TF-IDF text representation and cosine similarity to compare a search query with available member information. A similarity threshold helps determine whether a match is suitable. Results depend on the accuracy and completeness of the approved records.

5. Mobile Application

The documented interface includes a search input, search action, loading feedback, result details, error handling, and project information. The project is configured for Android builds through Expo Application Services (EAS). A successful final APK installation and end-to-end device test still need to be confirmed and evidenced.

6. Backend Testing

The recorded pytest run completed with 7 passed, 1 warning, and 0 failed. The warning concerned Starlette test-client integration with httpx. This records the backend automated tests only; it does not verify all mobile or deployment requirements.

7. Responsible Use and Limitations

• Member records should be verified before use.
• Text similarity can produce incorrect or incomplete matches.
• A similarity score is not proof of factual correctness.
• The API address must be appropriate for the deployment environment.
• Search results should be checked against approved information.
• Backend tests alone do not establish that the complete Android app works.

8. Collaboration and Evidence

The group should complete the contribution record using each member's actual work and evidence. Obtain real tester feedback, capture screenshots and the actual test output, and record the final Android build and installation checks. Do not submit placeholders as completed evidence.

9. Conclusion

AgriSoko Pocket combines a mobile search interface with an AI text-matching backend. Before final submission, verify the final application build, mobile-to-backend connectivity, assignment requirements, and supporting evidence.